Arun Dash

arundash@outlook.com | (623) 499-6900| linkedin.com/in/arun-dash

SUMMARY

Data Scientist / Data Engineer with 3+ years of experience building data pipelines, computer vision workflows, and biological data analysis tools. Strong background in Python, statistics, and bioinformatics with hands-on experience in ML model development, HPC workflows, and large-scale data processing.

EDUCATION

BSc. Bioinformatics with Molecular and Cellular Biology Specialization Tucson, AZ | Summer 2022
University of Arizona

WORK EXPERIENCE

BOEHRINGER INGELHEIM | Data Scientist March 2023 – October 2025

- Built reproducible Python data workflows that improved analysis turnaround time by 30% through automation and standardized pipelines.
- Created and annotated 10,000+ scientific images/videos for computer vision model training, supporting diverse use cases including kidney cell morphology and animal behavior tracking.
- Automated extraction of behavioral and spatial metrics from videos using Python, OpenCV, and NumPy, producing analytics used by downstream ML teams.
- Ran computational jobs on HPC clusters (SLURM, Bash) to train CV models, manage ETL-style data processing, and compile experiment outputs.
- Collaborated with cross-functional scientific teams to define data requirements, improve workflow reproducibility, and document analysis protocols.

GENOME SOLVER PROJECT | Undergraduate Research Assistant June 2021 – March 2023

- Enhanced Python BLAST automation pipeline, reducing runtime by 40% and improving downstream data parsing reliability.
- Utilized Pandas and web scraping tools to streamline data retrieval, processing, and formatting for comparative genomics studies.
- Refactored codebase for readability and accessibility, enabling use by high school students learning computational biology.

ACADEMIC PROJECTS

ISTA 335 FINAL PROJECT | Python, Pandas, Statsmodels Jan 2021 – May 2021

- Performed exploratory data analysis on global COVID-19 datasets, generating statistical visualizations using Pandas, Statsmodels, and Matplotlib.
- Applied web scraping and data cleaning techniques to construct a reproducible data processing pipeline.

GENE NETWORK RESEARCH | Berkeley-Madonna Aug 2019 – Dec 2019

- Modeled genetic networks using differential equations representing stochastic molecular interactions.
- Created simulations and visualizations illustrating gene expression oscillations and protein-level dynamics.

BRCA1 GENE EXPRESSION ANALYSIS | R, RNA-Seq Jan 2021 – May 2021

- Performed k-means clustering on RNA-Seq expression data in R to identify BRCA1-related phenotypic groupings.
- Conducted GO enrichment analysis to uncover pathways associated with breast cancer molecular signatures.

SKILLS

Languages: Python, SQL, Java, R, Bash
Data Engineering: Linux, HPC (SLURM), ETL Pipelines, Shell Scripting, Jupyter, Git
Soft Skills: Teamwork, Communication, Problem Solving
ML/Data: Pandas, NumPy, Scikit-Learn
Bioinformatics: BLAST, RNA-Seq, GO Analysis
Other: Microsoft Office, Data Visualization